Nitin Mittapally

868 682 7225 • mittapally.nitin@gmail.com • LinkedIn • GitHub • Portfolio • Hyderabad, India

SKILLS & COMPETENCIES

- ***Languages***: Python, Java, Scala, JavaScript, SQL, R, SAS
- ***GenAI & LLM*****:** LLMs, Hugging Face, Fine-tuning, Prompt Engineering, RAG Systems, LangChain, Single and Multi-Agentic AI Systems, Few-shot Learning, Vector Databases
- ***ML Frameworks*****:** PyTorch, TensorFlow, scikit-learn, Pandas, NumPy
- ***Platforms*****:** Google Cloud, AWS, Docker, Kubernetes, Jenkins CI/CD, GitHub Actions, Git
- ***Specializations:*** Natural Language Processing (NLP), Deep learning, Statistics, Machine Learning, Data Modeling, Feature engineering
- ***Databases:*** PostgreSQL, Big Query, Oracle DB, Vector DB (FAISS), MySQL DB, Mongo DB, SQLite
- ***Software Engineering***: Backend, Fast API, Spring boot, Django, Fronted, React, Streamlit

WORK EXPERIENCE

Charles Schwab — *California, USA*

Applied AI Engineer — *Dec 2025 – July 2026*

- Architected multi-Tenant organization-wide **Evals Platform** with Open Telemetry based ingestion, configurable sampling strategies, customizable evaluation metrics on GCP with VertexAI processing **~100K evals per hour**
- Built scalable **ETL pipelines** to ingest, summarize, and index daily financial news, market performance, and sector trends for all S&P 500 stocks using **Python, Airflow, and Google VertexAI**, creating reusable **knowledge repository**
- Developed a Portfolio Insights Generator service in Python that produces **personalized, context-aware investment insights** for Schwab customers based on **portfolio composition**, **market signals and daily P&L**
- **Engineered prompt strategies**, **evaluation frameworks**, and **safety guardrails** to ensure accurate, grounded, safe and compliant AI-generated insights, improving output reliability and consistency
- Conducted **load and stress testing of FastAPI** services to achieve **p99 latency under 3 seconds** at **150 Request/Sec**
- Designed and built React UI to **experiment with prompts, review model outputs**, **collect** human **feedback**, and visualize **evaluation metrics** through dashboards
- Optimized **prompt structure and token usage** to improve generated-content quality while reducing response latency by 15% and per-query cost by 25%

Fractal Analytics — *New York, USA*

Machine learning Engineer **(Client: Google)** — *Apr 2023 – Nov 2025*

- Developed and deployed large-scale **LLM App for feature extraction**, reducing 20 man-hours per user request, creating a centralized repository for generating latent transcript features for various Google hardware devices
- Designed and implemented large-scale end-to-end **BERT-based topic modeling pipeline**, uncovering trending issues and operational anomalies from chat transcripts, **driving proactive decision-making** leveraging hugging face and PyTorch
- Engineered **efficient and scalable foundational framework** for **prompt engineering,** equipping team of 20+ members to optimize workflows and enhance performance, reducing the 50+ man-hours per task
- Enhanced transcript filtering precision by integrating **semantic search** using FAISS **vector database**, improving data accessibility and reducing manual queries for context-based search
- **Fine-tuned BERT Model** for generating more accurate embeddings for Google Product customer transcripts using Huggingface
- Built **classification algorithms** to identify troubleshooting steps and **clustered** them, creating reusable documentation that increased self-service resolution rates by 25% using customer chat and phone transcripts leveraging scikit learn and PyTorch
- Conducted **gap analysis** for Google Help Center content to identify improvement areas for the **Pixel Tablet launch** based of Google Pixel Slate **search data** improving customer experience using transformers, scikit learn and PyTorch
- Extracted **actionable insights from customer reviews** of **Google Pixel Buds and Pixel Watch**, driving feature enhancements with competitor analysis of 500K+ reviews from various sources
- Leveraged **prompt engineering** and **few-shot learning** to improve Google Help Center resources and documentation for **six major product lines** by quantifying and analyzing educational transcript volumes

Data Scientist *Sep 2021 – Apr 2023*

- Implemented **BERTopic** to identify topics and subtopics in HR and IT support systems with 70% accuracy
- Developed **Data Processing Pipeline in Python** to clean global employee ticket corpus, remove corpus Stop-words, lemmatize and perform **Named Entity Recognition (NER)** to build datasets for topic modeling
- Consolidated Knowledge Base articles, reducing redundancy by **40%**; Scaled frequent articles to multiple languages
- Designed **Power BI Dashboard** showing topic trends across employee attributes and mapped them to employee journey
- Performed Data Pre-processing and **Feature Engineering** on 38 million records of **real-time retail transaction data**
- Built **Random Forest Regression Model** resulting in **16%** increase in accuracy for predicting the delivery time of products

DBS Bank *Hyderabad, India*
Application Developer *Sep 2019 – Dec 2020*

- Developed distributed data pipelines in **Java Spring Boot** for fetching, processing, and distributing Securities data from Bloomberg and Reuters to downstream banking and trading application.
- Designed end to end module to migrate from IBOR to SORA, that can handle **200 RPS** for on demand interest rate calculations.
- Built **Classification Model for Anomaly Detection** in Python, conserving **~30 hours** per week for accounting team, decreasing false positive violations **by ~28%** which occurred on daily basis when latest securities data came into system
- Coded Angular8 Business Process style UI for monitoring, correcting and approving security rates to downstream systems.
- Designed targeted advertising model using **logistic regression, random forest, and XGBoost** to identify the potential investors for mutual funds and term deposits, allowing **18% reduction in marketing cost**
- Created and automated monthly **Tableau dashboards** and sales reports, to monitor KPIs for stake holders

Oracle *Hyderabad, India*
Software Developer *Jan 2017 – Sep 2019*

- Engineered new and migrated highly performant **PL/SQL based Property Management System (Opera)** to Oracle Cloud
- Built 40 microservices and deployed to Kubernetes for handling core reservation functionality with response time below 1sec
- Simplified Opera's UI, to reduce operational time and improved customer experience by **30% reduction in clicks per task**
- Started an **NLP Chatbot** "OpBot", built on Oracle Cloud and Opera's core **APIs** enabling faster Kiosk Check-ins at resorts
- Delivered custom **automation testing framework** for web services and UI, saving **40%** of regression testing time in Python
- Redesigned **production CI/CD pipelines**, decreasing deployment time of micro-services to cloud by **~50%** on Jenkins

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business *Ohio, USA*
Master of Science, Major: Business Analytics and Administration *Dec 2021*

Osmania University, Vasavi College of Engineering *Hyderabad, India*
Bachelor of Engineering, Major: Electronics and Communication *May 2017*

PROJECTS AND RESEARCH PAPER IMPLEMENTATIONS

Attention is all you need *– Paper implementation [Github]*

- Implemented core components of the Transformer architecture from scratch in PyTorch, including positional encoding, multi-head attention, layer normalization, and encoder-decoder blocks
- Built modular design allowing for experimentation with different attention mechanisms and positional encoding strategies

Instruct GPT *– Paper Implementation [Github]*

- Followed OpenAI Research Paper Instruct GPT for instruction finetuning distilgpt2, to follow instructions and be helpful Agent
- Worked with Stanford Alpaca dataset with 50K records for instruction finetuning, using Hugging Face trainer API and PyTorch

TinyGPT *– Lightweight Transformer Implementation [Github]*

- Developed efficient transformer-based language model using 2 transformer layers with model size 21K parameters
- Trained on Tiny Shakespeare and achieved coherent English looking text with loss of 1.7 and 1.75 on training and test data
- Created modular architecture allowing for experimentation of KV Caching, distributed training and ONNX model generation

***DocuAnswer** – LangChain & Streamlit based Rag application for Q&A [Github]*

- Developed a Retrieval-Augmented Generation (RAG) system to process PDF files and answer user queries using FAISS for vector search and OpenAI's GPT-4o-mini for contextual responses
- Deployed seamless workflow with Streamlit frontend and Flask REST API for efficient user interaction

***Causal Language Modeling** – CodeGPT [Github]*

- Implemented a causal language model for python code generation based on CodeSearchNet with ~450K functions
- Created custom Byte Pair Encoder tokenizer with normalizer, pre tokenizer to handle python special characters
- Trained the GPT 2 architecture on with hugging face trainer, model was able to generate pythonic code

CERTIFICATIONS

- AI Agents Fundamentals – Hugging Face
- MCP Fundamentals – Hugging Face
- Generative AI with Large Language Models – Deep learning AI
- Getting started with Large Language Models – Analytics Vidhya
- Neural Networks and Deep Learning – Deep learning AI
- End to End Machine Learning with TensorFlow on Google Cloud – Google Cloud